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Image capturing optical lens assembly, image capturing device and electronic device

Abstract

This disclosure provides an image capturing optical lens assembly including, in order from an object side to an image side: a first lens element with refractive power having an object-side surface being convex in a paraxial region thereof a second lens element having positive refractive power; a third lens element with refractive power having an image-side surface being concave in a paraxial region thereof a fourth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein both surfaces thereof being aspheric; a fifth lens element with refractive power having an object-side surface being concave in a paraxial region thereof and a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein both surfaces thereof being aspheric, and the image-side surface having at least one convex shape in an off-axis region thereof.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/569,917 filed on Jan. 6, 2022, now approved, which is a divisional application of U.S. application Ser. No. 16/856,148 filed on Apr. 23, 2020, now U.S. Pat. No. 11,256,070, which is a continuation application of U.S. application Ser. No. 16/135,167 filed on Sep. 19, 2018, now U.S. Pat. No. 10,670,838, which is a continuation application of U.S. application Ser. No. 15/497,385 filed on Apr. 26, 2017, now U.S. Pat. No. 10,107,991, which is a continuation application of U.S. application Ser. No. 14/671,426 filed on Mar. 27, 2015, now U.S. Pat. No. 9,671,588 and claims priority to Taiwan Application Serial Number 103142323, filed Dec. 5, 2014, which is incorporated by reference herein in its entirety.

BACKGROUND

Technical Field

(1) The present disclosure relates to an image capturing optical lens assembly and an image capturing device, and more particularly, to an image capturing optical lens assembly and an image capturing device applicable to electronic devices.

Description of Related Art

(2) As personal electronic products have been becoming more and more compact, the internal components of the electronic products are also required to be smaller in size than before, resulting in an increasing demand for compact optical systems. In addition to the demand of miniaturization, the reduction of the pixel size of sensors in the advanced semiconductor manufacturing technologies has enabled optical systems to evolve toward the field of higher megapixels. Meanwhile, the popularity of smart phones and tablet computers greatly boost the need for compact optical systems featuring high image quality.

(3) A conventional compact optical system is usually configured to have a positive first lens element and a negative second lens element. However, the arrangement of greater refractive power for the first two lens elements at the light incident side usually increases the sensitivity of the system and adversely affects the manufacturing yield rate.

(4) Therefore, a need exists in the art for an optical system that features compact size, high image quality, improved manufacturing yield rate and appropriate sensitivity.

SUMMARY

(5) According to one aspect of the present disclosure, an image capturing optical lens assembly includes, in order from an object side to an image side: a first lens element with refractive power having an object-side surface being convex in a paraxial region thereof; a second lens element having positive refractive power; a third lens element with refractive power having an image-side surface being concave in a paraxial region thereof; a fourth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface thereof being aspheric; a fifth lens element with refractive power having an object-side surface being concave in a paraxial region thereof; and a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, and an object-side surface and the image-side surface thereof being aspheric, wherein the image-side surface having at least one convex shape in an off-axis region thereof; wherein the image capturing optical lens assembly has a total of six lens elements with refractive power and further includes a stop between an imaged object and the third lens element; wherein a focal length of the image capturing optical lens assembly is f , a focal length of the fifth lens element is f_5 , a focal length of the sixth lens element is f_6 , a focal length of the second lens element is f_2 , a focal

length of the third lens element is f_3 , a curvature radius of the object-side surface of the first lens element is R_1 , and the following conditions are satisfied:

$$1.50 < |f/f_5| + |f/f_6|;$$

$$2.20 < |f/f_2| + |f/f_3|; \text{ and}$$

$$|R_1/f| < 1.50.$$

(6) According to another aspect of the present disclosure, an image capturing optical lens assembly includes, in order from an object side to an image side: a first lens element with negative refractive power having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof; a second lens element having positive refractive power; a third lens element with negative refractive power having an image-side surface being concave in a paraxial region thereof; a fourth lens element with refractive power having an object-side surface and an image-side surface thereof being aspheric; a fifth lens element having refractive power; and a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface thereof being aspheric, and the image-side surface having at least one convex shape in an off-axis region thereof; wherein the image capturing optical lens assembly has a total of six lens elements with refractive power and further includes a stop between an imaged object and the third lens element; wherein a focal length of the image capturing optical lens assembly is f , a focal length of the fifth lens element is f_5 , a focal length of the sixth lens element is f_6 , a curvature radius of the object-side surface of the first lens element is R_1 , and the following conditions are satisfied:

$$1.50 < |f/f_5| + |f/f_6|; \text{ and}$$

$$|R_1/f| < 1.50.$$

(7) According to another aspect of the present disclosure, an image capturing optical lens assembly includes, in order from an object side to an image side: a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof; a second lens element having positive refractive power; a third lens element having refractive power; a fourth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface thereof being aspheric; a fifth lens element with refractive power having an object-side surface being concave in a paraxial region thereof; and a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface thereof being aspheric, and the image-side surface having at least one convex shape in an off-axis region thereof; wherein the image capturing optical lens assembly has a total of six lens elements with refractive power and further includes a stop between an imaged object and the third lens element; wherein a focal length of the image capturing optical lens assembly is f , a focal length of the second lens element is f_2 , a focal length of the third lens element is f_3 , and the following condition is satisfied:

$$2.20 < |f/f_2| + |f/f_3|.$$

(8) According to another aspect of the present disclosure, an image capturing device includes the aforementioned image capturing optical lens assembly and an image sensor.

(9) According to another aspect of the present disclosure, an electronic device includes the aforementioned image capturing device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure;

(2) FIG. 1B shows longitudinal spherical aberration curves, astigmatic field curves and a distortion

curve of the image capturing device according to the 1st embodiment;

(3) FIG. 2A is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure;

(4) FIG. 2B shows longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment;

(5) FIG. 3A is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure;

(6) FIG. 3B shows longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment;

(7) FIG. 4A is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure;

(8) FIG. 4B shows longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment;

(9) FIG. 5A is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure;

(10) FIG. 5B shows longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment;

(11) FIG. 6A is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure;

(12) FIG. 6B shows longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment;

(13) FIG. 7A is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure;

(14) FIG. 7B shows longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment;

(15) FIG. 8A is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure;

(16) FIG. 8B shows longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment;

(17) FIG. 9A is a schematic view of an image capturing device according to the 9th embodiment of the present disclosure;

(18) FIG. 9B shows longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 9th embodiment;

(19) FIG. 10A shows a smart phone with an image capturing device of the present disclosure installed therein;

(20) FIG. 10B shows a tablet personal computer with an image capturing device of the present disclosure installed therein; and

(21) FIG. 10C shows a wearable device with an image capturing device of the present disclosure installed therein.

DETAILED DESCRIPTION

(22) The present disclosure provides an image capturing optical lens assembly including, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element, wherein the image capturing optical lens assembly has a total of six lens elements with refractive power and further includes a stop, such as an aperture stop. The stop is between an imaged object and the third lens element.

(23) The first lens element may have negative refractive power so as to favorably enlarge the field of view of the image capturing optical lens assembly. Alternatively, the first lens element may have positive refractive power, so that it provides the image capturing optical lens assembly with part of refractive power as it needs to be so as to favorably reduce the total track length of the image

capturing optical lens assembly. The object-side surface of the first lens element is convex in a paraxial region thereof, and the image-side surface of the first lens element is concave in a paraxial region thereof so as to correct the astigmatism of the image capturing optical lens assembly.

(24) The second lens element may have positive refractive power, so that it is favorable for distributing the refractive power of the first lens element and thereby to favorably reduce the photosensitivity of the image capturing optical lens assembly. The object-side surface of the second lens element may be convex in a paraxial region thereof so as to enhance the capability of correcting the astigmatism.

(25) The third lens element may have negative refractive power, which together with the positive refractive power of the second lens element can mitigate the deflection of light incident to the image capturing optical lens assembly to control the aberration and to reduce the sensitivity of the image capturing optical lens assembly, thereby improving the manufacturing yield rate. The object-side surface of the third lens element may be concave in a paraxial region thereof, and the image-side surface of the third lens element is concave in a paraxial region thereof so as to favorably correct the aberration of the image capturing optical lens assembly. The object-side surface of the third lens element may have at least one inflection point so as to effectively correct the aberration of the off-axis field.

(26) The object-side surface of the fourth lens element may be convex in a paraxial region thereof, and the image-side surface of the fourth lens element is concave in a paraxial region thereof so as to effectively correct the astigmatism of the image capturing optical lens assembly and to effectively mitigate the variation in the shape of the fourth lens element, thereby facilitating the formation of the fourth lens element. The image-side surface of the fourth lens element has at least one convex shape in an off-axis region thereof so as to favorably correct the aberration of the off-axis field.

(27) The fifth lens element may have positive refractive power, so that it is favorable for balancing the distribution of the positive refractive power. The object-side surface of the fifth lens element may be concave in a paraxial region thereof, and the image-side surface of the fifth lens element may be convex in a paraxial region thereof so that the capability of correcting the astigmatism is enhanced.

(28) The sixth lens element may have negative refractive power, so that it is favorable for the distribution of the negative refractive power of the image capturing optical lens assembly. The image-side surface of the sixth lens element is concave in a paraxial region thereof; this allows the principal point of the image capturing optical lens assembly to be placed away from the image surface, thereby reducing the total track length of the image capturing optical lens assembly. The image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof so as to effectively correct the aberration of the off-axis field.

(29) When a focal length of the image capturing optical lens assembly is f , a focal length of the fifth lens element is f_5 , a focal length of the sixth lens element is f_6 , and the following condition is satisfied: $1.50 < |f/f_5| + 1/f_6|$, the distribution of the refractive power of the image capturing optical lens assembly is more balanced and thereby to reduce the sensitivity of the image capturing optical lens assembly. Preferably, the following condition is satisfied: $2.50 < |f/f_5| + |f/f_6| < 5.0$.

(30) When the focal length of the image capturing optical lens assembly is f , a focal length of the second lens element is f_2 , a focal length of the third lens element is f_3 , and the following condition is satisfied: $2.20 < |f/f_2| + |f/f_3|$, the deflection of light incident to the image capturing optical lens assembly can be mitigated to control the aberration and reduce the sensitivity of the image capturing optical lens assembly, thereby improving the manufacturing yield rate.

(31) When a curvature radius of the object-side surface of the first lens element is R_1 , the focal length of the image capturing optical lens assembly is f , and the following condition is satisfied: $|R_1/f| < 1.50$, it is favorable for reducing the sensitivity of the image capturing optical lens assembly. Preferably, the following condition is satisfied: $|R_1/f| < 0.75$.

(32) When a central thickness of the second lens element is CT_2 , an axial distance between the first lens element and the second lens element is T_{12} , an axial distance between the second lens element and the third lens element is T_{23} , and the following condition is satisfied: $2.4 < CT_2 / (T_{12} + T_{23}) < 5.0$, the thickness of the second lens element and the axial distances between the first lens element, the second lens element and the third lens element will be more appropriate. Such an arrangement facilitates controlling the thickness of the second lens element and preventing the total track length of the image capturing optical lens assembly from being too long, and this is favorable for assembling and manufacturing lens elements.

(33) When the focal length of the image capturing optical lens assembly is f , a curvature radius of the object-side surface of the second lens element is R_3 , and the following condition is satisfied: $1.40 < f / R_3$, the sensitivity of the image capturing optical lens assembly can be effectively corrected and this is favorable for manufacturing yield rate.

(34) When the focal length of the image capturing optical lens assembly is f , a curvature radius of the image-side surface of the sixth lens element is R_{12} , and the following condition is satisfied: $0 < R_{12} / f < 0.40$, the incident angle of the light projecting onto an image sensor from the image capturing optical lens assembly can be favorably suppressed to increase the photosensitivity of the image capturing optical lens assembly.

(35) When an Abbe number of the fifth lens element is V_5 , an Abbe number of the third lens element is V_3 , an Abbe number of the fourth lens element is V_4 , and the following condition is satisfied: $1.0 < V_5 / (V_3 + V_4) < 1.5$, the chromatic aberration in the image capturing optical lens assembly can be effectively corrected.

(36) When an f-number of the image capturing optical lens assembly is F_{no} , and the following condition is satisfied: $1.6 < F_{no} < 2.5$, the ambient illuminance of the image capturing optical lens assembly can be favorably increased.

(37) When a central thickness of the first lens element is CT_1 , a distance in parallel with an optical axis from an axial vertex on the image-side surface of the first lens element to a maximum effective radius position on the image-side surface of the first lens element is Sag_{12} , and the following condition is satisfied: $CT_1 / |Sag_{12}| < 2.5$, it is favorable for the reduction of the sensitivity of the image capturing optical lens assembly and the manufacturing yield rate.

(38) When a central thickness of the fifth lens element is CT_5 , a central thickness of the sixth lens element is CT_6 , and the following condition is satisfied: $0.80 < CT_5 / CT_6$, the thickness of the lens elements is more appropriate, and this is favorable for manufacturing and assembling lens elements.

(39) According to the image capturing optical lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic material. When the lens elements are made of glass material, the distribution of the refractive power of the image capturing optical lens assembly may be more flexible to design. When the lens elements are made of plastic material, the manufacturing cost can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric (ASP), since the aspheric surface of the lens element is easy to form a shape other than spherical surfaces so as to have more controllable variables for eliminating the aberration thereof and to further decrease the required number of the lens elements, the total track length of the image capturing optical lens assembly can be effectively reduced.

(40) According to the image capturing optical lens assembly of the present disclosure, the image capturing optical lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop.

(41) According to the image capturing optical lens assembly of the present disclosure, a stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the image capturing optical lens assembly and the image surface and thereby to improve the image-sensing efficiency of an image sensor, such as a CCD or CMOS sensor. A middle stop disposed between the first lens

element and the image surface is favorable for enlarging the field of view of the image capturing optical lens assembly and thereby to provide a wider field of view for the same.

(42) According to the image capturing optical lens assembly of the present disclosure, when the lens element has a convex surface and the region of convex shape is not defined, it indicates that the surface is convex in the paraxial region thereof; when the lens element has a concave surface and the region of concave shape is not defined, it indicates that the surface is concave in the paraxial region thereof. Likewise, when the region of refractive power or focal length of a lens element is not defined, it indicates that the region of refractive power or focal length of the lens element is in the paraxial region thereof.

(43) According to the image capturing optical lens assembly of the present disclosure, an image surface of the image capturing optical lens assembly, based on the corresponding image sensor, can be a plane or a curved surface with any curvature, especially a curved surface being concave facing towards the object side.

(44) The image capturing optical lens assembly of the present disclosure can be optionally applied to moving focus optical systems. According to the image capturing optical lens assembly of the present disclosure, the image capturing optical lens assembly features good correction capability and high image quality, and can be applied to 3D (three-dimensional) image capturing applications and electronic devices, such as digital cameras, mobile devices, digital tablets, smart TV, wireless monitoring device, motion sensing input device, driving recording system, rear view camera system, and wearable devices.

(45) According to the present disclosure, an image capturing device includes the aforementioned image capturing optical lens assembly and an image sensor, wherein the image sensor is disposed on or near an image surface of the image capturing optical lens assembly. Therefore, the design of the image capturing optical lens assembly enables the image capturing device to achieve the best image quality. Preferably, the image capturing device can further include a barrel member, a holding member or a combination thereof.

(46) Referring to FIG. 10A, FIG. 10B and FIG. 10C, an image capturing device **1001** may be installed in an electronic device, including, but not limited to, a smart phone **1010**, a tablet personal computer **1020** or a wearable device **1030**. The three exemplary figures of different kinds of electronic devices are only exemplary for showing the image capturing device of present disclosure installed in an electronic device and are not limited thereto. Preferably, the electronic device can further include a control unit, a display unit, a storage unit, a random access memory unit (RAM) or a combination thereof.

(47) According to the above description of the present disclosure, the following 1st-9th specific embodiments are provided for further explanation.

1st Embodiment

(48) FIG. 1A is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure. FIG. 1B shows, in order from left to right, longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment.

(49) In FIG. 1A, the image capturing device includes an image capturing optical lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor **190**. The image capturing optical lens assembly includes, in order from an object side to an image side, a first lens element **110**, a second lens element **120**, a third lens element **130**, a fourth lens element **140**, a fifth lens element **150**, and a sixth lens element **160**, wherein the image capturing optical lens assembly has a total of six lens elements (**110-160**) with refractive power.

(50) The first lens element **110** with positive refractive power has an object-side surface **111** being convex in a paraxial region thereof and an image-side surface **112** being concave in a paraxial region thereof, which are both aspheric, and the first lens element **110** is made of plastic material.

(51) The second lens element **120** with positive refractive power has an object-side surface **121**

being convex in a paraxial region thereof and an image-side surface **122** being convex in a paraxial region thereof, which are both aspheric, and the second lens element **120** is made of plastic material.

(52) The third lens element **130** with negative refractive power has an object-side surface **131** being concave in a paraxial region thereof and an image-side surface **132** being concave in a paraxial region thereof, which are both aspheric, and the third lens element **130** is made of plastic material. Furthermore, the object-side surface **131** of the third lens element **130** has at least one inflection point.

(53) The fourth lens element **140** with negative refractive power has an object-side surface **141** being convex in a paraxial region thereof and an image-side surface **142** being concave in a paraxial region thereof, which are both aspheric, and the fourth lens element **140** is made of plastic material. Furthermore, the image-side surface **142** of the fourth lens element **140** has at least one convex shape in an off-axis region thereof.

(54) The fifth lens element **150** with positive refractive power has an object-side surface **151** being concave in a paraxial region thereof and an image-side surface **152** being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element **150** is made of plastic material.

(55) The sixth lens element **160** with negative refractive power has an object-side surface **161** being convex in a paraxial region thereof and an image-side surface **162** being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element **160** is made of plastic material. Furthermore, the image-side surface **162** of the sixth lens element **160** has at least one convex shape in an off-axis region thereof.

(56) The image capturing optical lens assembly further includes an aperture stop **100** and an IR-cut filter **170**. The aperture stop **100** is disposed between an imaged object and the first lens element **110**. The IR-cut filter **170** is made of glass and located between the sixth lens element **160** and an image surface **180**, and will not affect the focal length of the image capturing optical lens assembly.

(57) The image sensor **190** is disposed on or near the image surface **180** of the image capturing optical lens assembly.

(58) The equation of the aspheric surface profiles is expressed as follows:

$$(59) X(Y) = (Y^2 / R) / (1 + \sqrt{1 - (1 + k) (Y / R)^2}) + \sum_i A_i (Y^i),$$

where, X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis; Y is the vertical distance from the point on the aspheric surface to the optical axis; R is the curvature radius; k is the conic coefficient; and A_i is the i-th aspheric coefficient.

(60) In the first embodiment, a focal length of the image capturing optical lens assembly is f, an f-number of the image capturing optical lens assembly is F_{no} , half of maximal field of view of the image capturing optical lens assembly is HFOV, and these parameters have the following values: $f=3.78$ mm; $F_{no}=1.90$; and HFOV=38.5 degrees.

(61) In the first embodiment, an Abbe number of the fifth lens element **150** is V_5 , an Abbe number of the third lens element **130** is V_3 , an Abbe number of the fourth lens element **140** is V_4 , and they satisfy the condition: $V_5/(V_3+V_4)=1.19$.

(62) In the first embodiment, a central thickness of the second lens element **120** is CT_2 , an axial distance between the first lens element **110** and the second lens element **120** is T_{12} , an axial distance between the second lens element **120** and the third lens element **130** is T_{23} , and they satisfy the condition: $CT_2/(T_{12}+T_{23})=2.70$.

(63) In the first embodiment, a central thickness of the fifth lens element **150** is CT_5 , a central thickness of the sixth lens element **160** is CT_6 , and they satisfy the condition: $CT_5/CT_6=1.99$.

(64) In the first embodiment, a central thickness of the first lens element **110** is CT_1 , a distance in parallel with an optical axis from an axial vertex on the image-side surface **112** of the first lens element **110** to a maximum effective radius position on the image-side surface **112** of the first lens

element **110** is Sag12, and they satisfy the condition: $CT1/|Sag12|=1.47$.

(65) In the first embodiment, a curvature radius of the object-side surface **111** of the first lens element **110** is R1, a focal length of the image capturing optical lens assembly is f, and they satisfy the condition: $|R1/f|=0.46$.

(66) In the first embodiment, the focal length of the image capturing optical lens assembly is f, a curvature radius of the object-side surface **121** of the second lens element **120** is R3, and they satisfy the condition: $f/R3=1.77$.

(67) In the first embodiment, a curvature radius of the image-side surface **162** of the sixth lens element **160** is R12, the focal length of the image capturing optical lens assembly is f, and they satisfy the condition: $R12/f=0.20$.

(68) In the first embodiment, the focal length of the image capturing optical lens assembly is f, a focal length of the second lens element **120** is f2, a focal length of the third lens element **130** is f3, and they satisfy the condition: $|f/f2|+|f/f3|=2.28$.

(69) In the first embodiment, the focal length of the image capturing optical lens assembly is f, a focal length of the fifth lens element **150** is f5, a focal length of the sixth lens element **160** is f6, and they satisfy the condition: $|f/f5|+|f/f6|=3.61$.

(70) The detailed optical data of the first embodiment are shown in TABLE 1, and the aspheric surface data are shown in TABLE 2, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximal field of view.

(71) TABLE-US-00001 TABLE 1 (Embodiment 1) f = 3.78 mm, Fno = 1.90, HFOV = 38.5 deg.

Focal Surface #	Curvature	Radius	Thickness	Material	Index	Abbe#	Length	0 Object	Plano	Infinity
1 Ape. Stop	Plano	-0.247	2	Lens 1	1.755	ASP	0.300	Plastic	1.544	55.9
419.65	3	1.662	ASP	0.151	4	Lens 2	2.133	ASP	0.594	Plastic
1.544	55.9	2.78	5	-4.662	ASP	0.069	6	Lens 3	-38.635	ASP
0.265	Plastic	1.639	23.5	-4.12	7	2.830	ASP	0.293	8	Lens 4
3.671	ASP	0.310	Plastic	1.633	23.4	-843.52	9	3.527	ASP	0.330
10	Lens 5	-12.309	ASP	0.899	Plastic	1.544	55.9	2.02	11	-1.035
ASP	0.050	12	Lens 6	2.468	ASP	0.451	Plastic	1.544	55.9	-2.18
13	0.750	ASP	0.800	14	IR-cut filter	Plano	0.210	Glass	1.517	64.2
15	Plano	0.470	16	Image Surface	Plano	—	Note: Reference Wavelength is d-line 587.6 nm			

(72) TABLE-US-00002 TABLE 2 Aspheric Coefficients Surface # 2 3 4 5 6 7 k= -5.8403E-01

-1.7275E+00	-7.6830E+00	-9.0000E+01	8.9992E+01	3.1605E+00	A4= -4.1244E-02		
-8.7529E-02	1.2639E-02	-2.1623E-03	1.3639E-01	3.8318E-03	A6= 8.8742E-03		
4.0028E-02	-5.2013E-02	-1.9764E-01	-4.2352E-01	-1.1664E-01	A8= 6.8214E-03		
-3.6180E-02	-1.4139E-02	1.7852E-01	4.5265E-01	1.0855E-01	A10= -1.5515E-02		
9.2136E-03	-5.4540E-02	-2.1953E-01	-5.8999E-02	A12= 4.0773E-02	1.1469E-02		
Surface #	8	9	10	11	12	13	k= -5.9209E+01
-1.8336E+01	-3.3470E+00	-5.5845E+00	-2.4679E-01	-4.3394E+00	A4= -2.1805E-02		
-7.9011E-02	7.7778E-02	-5.0173E-02	-2.1999E-01	-9.0041E-02	A6= -3.2370E-02		
3.1538E-02	-1.2567E-01	1.4292E-02	6.7649E-02	3.2227E-02	A8= 4.3459E-03		
-2.1342E-02	1.3643E-01	9.6967E-03	-1.0551E-02	-8.5109E-03	A10= 1.3091E-02		
-1.4490E-02	-9.6542E-02	-9.4547E-04	9.7036E-04	1.3883E-03	A12= -5.2554E-03		
2.1447E-02	3.6315E-02	-1.0958E-03	-7.6596E-05	-1.3093E-04	A14= -5.3783E-03		
-5.4872E-03	1.8051E-04	4.1836E-06	5.5499E-06				

2nd Embodiment

(73) FIG. 2A is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure. FIG. 2B shows, in order from left to right, longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment.

(74) In FIG. 2A, the image capturing device includes an image capturing optical lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor **290**. The image capturing optical lens assembly includes, in order from an object side to an image side, a first lens element **210**, a second lens element **220**, a third lens element **230**, a fourth lens element **240**, a fifth lens

element **250**, and a sixth lens element **260**, wherein the image capturing optical lens assembly has a total of six lens elements (**210-260**) with refractive power.

(75) The first lens element **210** with negative refractive power has an object-side surface **211** being convex in a paraxial region thereof and an image-side surface **212** being concave in a paraxial region thereof, which are both aspheric, and the first lens element **210** is made of plastic material.

(76) The second lens element **220** with positive refractive power has an object-side surface **221** being convex in a paraxial region thereof and an image-side surface **222** being convex in a paraxial region thereof, which are both aspheric, and the second lens element **220** is made of plastic material.

(77) The third lens element **230** with negative refractive power has an object-side surface **231** being concave in a paraxial region thereof and an image-side surface **232** being concave in a paraxial region thereof, which are both aspheric, and the third lens element **230** is made of plastic material. Furthermore, the object-side surface **231** of the third lens element **230** has at least one inflection point.

(78) The fourth lens element **240** with positive refractive power has an object-side surface **241** being convex in a paraxial region thereof and an image-side surface **242** being concave in a paraxial region thereof, which are both aspheric, and the fourth lens element **240** is made of plastic material. Furthermore, the image-side surface **242** of the fourth lens element **240** has at least one convex shape in an off-axis region thereof.

(79) The fifth lens element **250** with positive refractive power has an object-side surface **251** being convex in a paraxial region thereof and an image-side surface **252** being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element **250** is made of plastic material.

(80) The sixth lens element **260** with negative refractive power has an object-side surface **261** being convex in a paraxial region thereof and an image-side surface **262** being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element **260** is made of plastic material. Furthermore, the image-side surface **262** of the sixth lens element **260** has at least one convex shape in an off-axis region thereof.

(81) The image capturing optical lens assembly further includes an aperture stop **200** and an IR-cut filter **270**. The aperture stop **200** is disposed between an imaged object and the first lens element **210**. The IR-cut filter **270** is made of glass and located between the sixth lens element **260** and an image surface **280**, and will not affect the focal length of the image capturing optical lens assembly.

(82) The image sensor **290** is disposed on or near the image surface **280** of the image capturing optical lens assembly.

(83) The detailed optical data of the second embodiment are shown in TABLE 3, and the aspheric surface data are shown in TABLE 4, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximal field of view.

(84) TABLE-US-00003 TABLE 3 (Embodiment 2) $f = 3.85$ mm, $Fno = 2.03$, HFOV = 38.0 deg.

Focal Surface #	Curvature Radius	Thickness	Material	Index	Abbe #	Length	0	Object	Plano	Infinity
1 Ape. Stop	Plano	-0.205	2 Lens 1	1.865	ASP	0.300	Plastic	1.639	23.5	-17.00
3 1.492	ASP	0.125	4 Lens 2	1.946	ASP	0.746	Plastic	1.544	55.9	2.58
5 -4.338	ASP	0.082	6 Lens 3	-43.214	ASP	0.257	Plastic	1.639	23.5	-4.50
7 3.089	ASP	0.286	8 Lens 4	4.419	ASP	0.294	Plastic	1.639	23.5	43.61
9 5.115	ASP	0.576	10 Lens 5	23.394	ASP	0.631	Plastic	1.544	55.9	2.32
11 -1.323	ASP	0.050	12 Lens 6	2.113	ASP	0.437	Plastic	1.535	55.7	-2.42
13 0.744	ASP	0.800	14 IR-cut filter	Plano	0.210	Glass	1.517	64.2	15 Plano	0.468
16 Image	Plano	—	Surface Note:	Reference wavelength is d-line	587.6 nm					

(85) TABLE-US-00004 TABLE 4 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k = -9.5006E-01$

$-1.4696E+00$ $-5.1972E+00$ $-9.0000E+01$ $1.8649E+01$ $2.8260E+00$ $A4 = -4.8463E-02$

$-7.9417E-02$ $2.6637E-02$ $-1.8368E-02$ $1.1132E-01$ $-1.1792E-02$ $A6 = 1.7148E-02$ $3.7001E-02$

$-3.8254E-02$ $-1.7579E-01$ $-4.3697E-01$ $-1.2813E-01$ $A8 = -5.3900E-03$ $-1.5693E-02$

$9.7414E-03$ $1.6829E-01$ $4.4501E-01$ $1.2110E-01$ $A10 = -3.0618E-03$ $-3.9222E-03$

-5.1187E-02 -1.8095E-01 -4.9657E-02 A12= 2.5264E-02 6.5309E-03 Surface # 8 9 10 11 12
 13 k= -5.9209E+01 -1.8336E+01 -3.3470E+00 -8.3516E+00 -2.6657E-01 -3.7283E+00 A4=
 -8.4909E-02 -1.5264E-01 9.4160E-02 4.6088E-02 -2.4725E-01 -1.2284E-01 A6=
 5.6241E-02 8.3547E-02 -1.2263E-01 -8.1544E-02 5.4175E-03 4.7993E-02 A8= -9.6690E-02
 -6.4550E-02 9.9072E-02 7.2034E-02 5.1453E-02 -1.2048E-02 A10= 7.3382E-02 1.3063E-02
 -5.3443E-02 -2.5383E-02 -2.0237E-02 1.8540E-03 A12= -1.7548E-02 1.6176E-02
 1.5774E-02 3.5764E-03 3.0896E-03 -1.6330E-04 A14= -5.6694E-03 -2.0349E-03
 -1.5400E-04 -1.7312E-04 6.2775E-06

(86) In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in Table 5 below are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

(87) Moreover, these parameters can be calculated from Table 3 and Table 4 and satisfy the conditions stated in Table 5.

(88) TABLE-US-00005 TABLE 5 2.sup.nd Embodiment f [mm] 3.85 Fno 2.03 HFOV [deg.] 38.0
 V5/(V3 + V4) 1.19 CT2/(T12 + T23) 3.60 CT5/CT6 1.44 CT1/|Sag12| 1.11 |R1/f| 0.48 f/R3 1.98
 R12/f 0.19 |f/f2| + |f/f3| 2.35 |f/f5| + |f/f6| 3.25

3rd Embodiment

(89) FIG. 3A is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure. FIG. 3B shows, in order from left to right, longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment.

(90) In FIG. 3A, the image capturing device includes an image capturing optical lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 390. The image capturing optical lens assembly includes, in order from an object side to an image side, a first lens element 310, a second lens element 320, a third lens element 330, a fourth lens element 340, a fifth lens element 350, and a sixth lens element 360, wherein the image capturing optical lens assembly has a total of six lens elements (310-360) with refractive power.

(91) The first lens element 310 with positive refractive power has an object-side surface 311 being convex in a paraxial region thereof and an image-side surface 312 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 310 is made of plastic material.

(92) The second lens element 320 with positive refractive power has an object-side surface 321 being convex in a paraxial region thereof and an image-side surface 322 being convex in a paraxial region thereof, which are both aspheric, and the second lens element 320 is made of glass.

(93) The third lens element 330 with negative refractive power has an object-side surface 331 being concave in a paraxial region thereof and an image-side surface 332 being concave in a paraxial region thereof, which are both aspheric, and the third lens element 330 is made of plastic material. Furthermore, the object-side surface 331 of the third lens element 330 has at least one inflection point.

(94) The fourth lens element 340 with negative refractive power has an object-side surface 341 being convex in a paraxial region thereof and an image-side surface 342 being concave in a paraxial region thereof, which are both aspheric, and the fourth lens element 340 is made of plastic material. Furthermore, the image-side surface 342 of the fourth lens element 340 has at least one convex shape in an off-axis region thereof.

(95) The fifth lens element 350 with positive refractive power has an object-side surface 351 being concave in a paraxial region thereof and an image-side surface 352 being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element 350 is made of plastic material.

(96) The sixth lens element 360 with negative refractive power has an object-side surface 361 being convex in a paraxial region thereof and an image-side surface 362 being concave in a

paraxial region thereof, which are both aspheric, and the sixth lens element **360** is made of plastic material. Furthermore, the image-side surface **362** of the sixth lens element **360** has at least one convex shape in an off-axis region thereof.

(97) The image capturing optical lens assembly further includes an aperture stop **300** and an IR-cut filter **370**. The aperture stop **300** is disposed between an imaged object and the first lens element **310**. The IR-cut filter **370** is made of glass and located between the sixth lens element **360** and an image surface **380**, and will not affect the focal length of the image capturing optical lens assembly.

(98) The image sensor **390** is disposed on or near the image surface **380** of the image capturing optical lens assembly.

(99) The detailed optical data of the third embodiment are shown in TABLE 6, and the aspheric surface data are shown in TABLE 7, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximal field of view.

(100) TABLE-US-00006 TABLE 6 (Embodiment 3) $f = 4.00$ mm, $Fno = 2.20$, $HFOV = 37.0$ deg.
Curvature Focal Surface # Radius Thickness Material Index Abbe # Length 0 Object Plano Infinity
1 Ape. Stop Plano -0.234 2 Lens 1 1.641 ASP 0.364 Plastic 1.544 55.9 220.35 3 1.533 ASP 0.171 4
Lens 2 2.242 ASP 0.632 Glass 1.592 60.6 2.59 5 -4.344 ASP 0.072 6 Lens 3 -97.807 ASP 0.265
Plastic 1.650 21.5 -4.59 7 3.084 ASP 0.510 8 Lens 4 8.562 ASP 0.310 Plastic 1.650 21.5 -51.98 9
 6.735 ASP 0.319 10 Lens 5 -3.775 ASP 0.562 Plastic 1.544 55.9 1.95 11 -0.870 ASP 0.050 12
Lens 6 20.329 ASP 0.498 Plastic 1.544 55.9 -1.92 13 0.987 ASP 0.800 14 IR-cut filter Plano 0.300
Glass 1.517 64.2 15 Plano 0.330 16 Image Surface Plano — Note: Reference wavelength is d-line
 587.6 nm

(101) TABLE-US-00007 TABLE 7 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k = -2.3625E-01$
 $-5.1323E-01$ $-6.3496E+00$ $-8.5546E+01$ $9.0000E+01$ $4.7426E+00$ $A4 = -2.9819E-02$
 $-4.2192E-02$ $4.4302E-02$ $9.0303E-04$ $1.4707E-01$ $4.7795E-03$ $A6 = -1.9524E-02$ $-3.7052E-02$
 $-5.3859E-02$ $-1.9805E-01$ $-4.2318E-01$ $-1.1289E-01$ $A8 = 1.1402E-02$ $3.0115E-03$
 $-2.6940E-02$ $1.9648E-01$ $4.9123E-01$ $1.1398E-01$ $A10 = -9.6681E-03$ $2.0339E-02$
 $-7.0527E-02$ $-2.2873E-01$ $-3.9910E-02$ $A12 = 2.8300E-02$ $-4.5156E-03$ Surface # 8 9 10 11 12
13 $k = -9.0000E+01$ $-1.8336E+01$ $4.1711E-01$ $-3.1586E+00$ $3.1571E+00$ $-6.1836E+00$ $A4 =$
 $-1.1006E-01$ $-2.1050E-01$ $-2.7733E-01$ $-8.8569E-02$ $-2.8139E-02$ $-6.3400E-02$ $A6 =$
 $3.7911E-02$ $2.1305E-01$ $1.4087E-01$ $-1.1462E-01$ $1.9850E-02$ $2.4929E-02$ $A8 = -2.5110E-02$
 $-2.2483E-01$ $1.9670E-01$ $3.0457E-01$ $-1.2431E-02$ $-6.9183E-03$ $A10 = -3.2822E-02$
 $1.6220E-01$ $-2.2965E-01$ $-2.0229E-01$ $3.6192E-03$ $1.0336E-03$ $A12 = 1.7673E-02$
 $-7.9151E-02$ $8.3773E-02$ $5.6359E-02$ $-4.7701E-04$ $-8.1559E-05$ $A14 = 1.7180E-02$
 $-1.1045E-02$ $-5.8255E-03$ $2.3696E-05$ $2.8318E-06$

(102) In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in Table 8 below are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

(103) Moreover, these parameters can be calculated from Table 6 and Table 7 and satisfy the conditions stated in Table 8.

(104) TABLE-US-00008 TABLE 8 3.sup.rd Embodiment f [mm] 4.00 Fno 2.20 $HFOV$ [deg.] 37.0
 $V5/(V3 + V4)$ 1.30 $CT2/(T12 + T23)$ 2.60 $CT5/CT6$ 1.13 $CT1/|Sag12|$ 1.40 $|R1/f|$ 0.41 $f/R3$ 1.78
 $R12/f$ 0.25 $|f/f2| + |f/f3|$ 2.42 $|f/f5| + |f/f6|$ 4.13

4th Embodiment

(105) FIG. 4A is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure. FIG. 4B shows, in order from left to right, longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment.

(106) In FIG. 4A, the image capturing device includes an image capturing optical lens assembly

(not otherwise herein labeled) of the present disclosure and an image sensor **490**. The image capturing optical lens assembly includes, in order from an object side to an image side, a first lens element **410**, a second lens element **420**, a third lens element **430**, a fourth lens element **440**, a fifth lens element **450**, and a sixth lens element **460**, wherein the image capturing optical lens assembly has a total of six lens elements (**410-460**) with refractive power.

(107) The first lens element **410** with negative refractive power has an object-side surface **411** being convex in a paraxial region thereof and an image-side surface **412** being concave in a paraxial region thereof, which are both aspheric, and the first lens element **410** is made of plastic material.

(108) The second lens element **420** with positive refractive power has an object-side surface **421** being convex in a paraxial region thereof and an image-side surface **422** being convex in a paraxial region thereof, which are both aspheric, and the second lens element **420** is made of plastic material.

(109) The third lens element **430** with negative refractive power has an object-side surface **431** being concave in a paraxial region thereof and an image-side surface **432** being concave in a paraxial region thereof, which are both aspheric, and the third lens element **430** is made of plastic material. Furthermore, the object-side surface **431** of the third lens element **430** has at least one inflection point.

(110) The fourth lens element **440** with positive refractive power has an object-side surface **441** being concave in a paraxial region thereof and an image-side surface **442** being convex in a paraxial region thereof, which are both aspheric, and the fourth lens element **440** is made of plastic material.

(111) The fifth lens element **450** with positive refractive power has an object-side surface **451** being concave in a paraxial region thereof and an image-side surface **452** being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element **450** is made of plastic material.

(112) The sixth lens element **460** with negative refractive power has an object-side surface **461** being concave in a paraxial region thereof and an image-side surface **462** being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element **460** is made of plastic material. Furthermore, the image-side surface **462** of the sixth lens element **460** has at least one convex shape in an off-axis region thereof.

(113) The image capturing optical lens assembly further includes an aperture stop **400** and an IR-cut filter **470**. The aperture stop **400** is disposed between the first lens element **410** and the second lens element **420**. The IR-cut filter **470** is made of glass and located between the sixth lens element **460** and an image surface **480**, and will not affect the focal length of the image capturing optical lens assembly.

(114) The image sensor **490** is disposed on or near the image surface **480** of the image capturing optical lens assembly.

(115) The detailed optical data of the fourth embodiment are shown in TABLE 9, and the aspheric surface data are shown in TABLE 10, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximal field of view.

(116) TABLE-US-00009 TABLE 9 (Embodiment 4) $f = 4.05$ mm, $Fno = 2.35$, $HFOV = 37.4$ deg.

Curvature	Focal	Surface #	Radius	Thickness	Material	Index	Abbe #	Length	0	Object	Plano	Infinity
1	Lens 1	1.983	ASP	0.483	Plastic	1.639	23.5	-28.07	2	1.616	ASP	0.216
3	Ape. Stop	Plano	-0.054	4	Lens 2	1.603	ASP	0.635	Plastic	1.544	55.9	2.16
5	-3.774	ASP	0.050	6	Lens 3	-8.904	ASP	0.265	Plastic	1.639	23.5	-3.79
7	3.364	ASP	0.476	8	Lens 4	-76.136	ASP	0.549	Plastic	1.639	23.5	19.86
9	-10.908	ASP	0.232	10	Lens 5	-3.216	ASP	0.450	Plastic	1.560	48.0	2.21
11	-0.939	ASP	0.110	12	Lens 6	-10.240	ASP	0.457	Plastic	1.535	55.7	-1.87
13	1.128	ASP	0.700	14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	15	Plano
0.385	16	Image Surface	Plano	—	Note: Reference wavelength is d-line 587.6 nm							

(117) TABLE-US-00010 TABLE 10 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k = -5.7523E-01$

-2.1481E+00 -5.1057E+00 -2.0516E+00 -7.1165E+01 3.7642E+00 A4= -3.7779E-02
 -8.5995E-02 3.7763E-02 6.4233E-02 6.5837E-02 -9.9342E-03 A6= 9.2238E-03 2.3775E-02
 -7.6710E-02 -3.5241E-01 -3.2167E-01 -9.6870E-03 A8= -1.2136E-02 -3.6991E-02
 -8.6875E-02 2.6822E-01 4.1527E-01 -1.8667E-02 A10= 1.0602E-02 3.6436E-02 3.9520E-02
 -7.1191E-02 -1.5582E-01 2.2647E-01 A12= -3.5836E-03 4.6081E-07 -1.7005E-07
 -3.7222E-05 9.5053E-03 -3.7548E-01 A14 -1.6420E-06 9.3737E-10 -2.4239E-08
 2.6821E-08 3.5253E-05 2.4673E-01 A16 -5.0985E-07 3.2671E-10 -3.5042E-09 5.0416E-09
 6.9823E-09 -5.7653E-02 Surface # 8 9 10 11 12 13 k= 8.9919E+01 -2.0465E+01 2.9303E+00
 -2.7882E+00 1.0183E+01 -5.4912E+00 A4= 1.5752E-02 5.1742E-02 -1.5154E-01
 -1.0657E-01 -3.9771E-02 -1.0047E-01 A6= -3.5756E-01 -3.6773E-01 1.1369E-01
 6.2159E-02 7.8334E-02 6.2075E-02 A8= 9.4123E-01 5.9434E-01 -1.4877E-01 3.7236E-02
 -7.7804E-02 -2.8959E-02 A10= -1.5526E+00 -6.0723E-01 2.3795E-01 -4.4330E-02
 3.5295E-02 8.2981E-03 A12= 1.4931E+00 4.0608E-01 -1.5071E-01 1.8291E-02 -8.0761E-03
 -1.4363E-03 A14= -7.7570E-01 -1.5902E-01 3.3077E-02 -3.7920E-03 9.2581E-04
 1.3625E-04 A16= 1.6501E-01 2.6042E-02 -9.6229E-04 3.2310E-04 -4.2687E-05
 -5.3431E-06

(118) In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in Table 11 below are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

(119) Moreover, these parameters can be calculated from Table 9 and Table 10 and satisfy the conditions stated in Table 11.

(120) TABLE-US-00011 TABLE 11 4.sup.th Embodiment f [mm] 4.05 Fno 2.35 HFOV [deg.] 37.4 V5/(V3 + V4) 1.02 CT2/(T12 + T23) 3.00 CT5/CT6 0.98 CT1/|Sag12| 2.91 |R1/f| 0.49 f/R3 2.53 R12/f 0.28 |f/f2| + |f/f3| 2.94 |f/f5| + |f/f6| 4.00

5th Embodiment

(121) FIG. 5A is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure. FIG. 5B shows, in order from left to right, longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment.

(122) In FIG. 5A, the image capturing device includes an image capturing optical lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 590. The image capturing optical lens assembly includes, in order from an object side to an image side, a first lens element 510, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, and a sixth lens element 560, wherein the image capturing optical lens assembly has a total of six lens elements (510-560) with refractive power.

(123) The first lens element 510 with negative refractive power has an object-side surface 511 being convex in a paraxial region thereof and an image-side surface 512 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 510 is made of plastic material.

(124) The second lens element 520 with positive refractive power has an object-side surface 521 being convex in a paraxial region thereof and an image-side surface 522 being convex in a paraxial region thereof, which are both aspheric, and the second lens element 520 is made of plastic material.

(125) The third lens element 530 with negative refractive power has an object-side surface 531 being concave in a paraxial region thereof and an image-side surface 532 being concave in a paraxial region thereof, which are both aspheric, and the third lens element 530 is made of plastic material. Furthermore, the object-side surface 531 of the third lens element 530 has at least one inflection point.

(126) The fourth lens element **540** with positive refractive power has an object-side surface **541** being convex in a paraxial region thereof and an image-side surface **542** being concave in a paraxial region thereof, which are both aspheric, and the fourth lens element **540** is made of plastic material. Furthermore, the image-side surface **542** of the fourth lens element **540** has at least one convex shape in an off-axis region thereof.

(127) The fifth lens element **550** with positive refractive power has an object-side surface **551** being concave in a paraxial region thereof and an image-side surface **552** being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element **550** is made of plastic material.

(128) The sixth lens element **560** with negative refractive power has an object-side surface **561** being convex in a paraxial region thereof and an image-side surface **562** being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element **560** is made of plastic material. Furthermore, the image-side surface **562** of the sixth lens element **560** has at least one convex shape in an off-axis region thereof.

(129) The image capturing optical lens assembly further includes an aperture stop **500** and an IR-cut filter **570**. The aperture stop **500** is disposed between an imaged object and the first lens element **510**. The IR-cut filter **570** is made of glass and located between the sixth lens element **560** and an image surface **580**, and will not affect the focal length of the image capturing optical lens assembly.

(130) The image sensor **590** is disposed on or near the image surface **580** of the image capturing optical lens assembly.

(131) The detailed optical data of the fifth embodiment are shown in TABLE 12, and the aspheric surface data are shown in TABLE 13, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximal field of view.

(132) TABLE-US-00012 TABLE 12 (Embodiment 5) $f = 4.13$ mm, $Fno = 2.20$, HFOV = 36.4 deg.
Curvature Focal Surface # Radius Thickness Material Index Abbe # Length 0 Object Plano Infinity
1 Ape. Stop Plano -0.255 2 Lens 1 1.500 ASP 0.300 Plastic 1.544 55.9 -166.89 3 1.371 ASP 0.151
4 Lens 2 1.865 ASP 0.612 Plastic 1.544 55.9 2.35 5 -3.576 ASP 0.050 6 Lens 3 -8.368 ASP 0.265
Plastic 1.639 23.5 -3.62 7 3.231 ASP 0.426 8 Lens 4 2.892 ASP 0.319 Plastic 1.639 23.5 27.68 9
3.308 ASP 0.450 10 Lens 5 -1.768 ASP 0.565 Plastic 1.544 55.9 2.15 11 -0.784 ASP 0.050 12
Lens 6 5.648 ASP 0.445 Plastic 1.544 55.9 -2.15 13 0.942 ASP 0.800 14 IR-cut filter Plano 0.300
Glass 1.517 64.2 15 Plano 0.457 16 Image Surface Plano — Note: Reference wavelength is d-line
587.6 nm

(133) TABLE-US-00013 TABLE 13 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k = -5.5896E-01$
 $-1.4494E+00$ $-6.3163E+00$ $-5.6354E+01$ $-4.3569E+01$ $5.3797E+00$ $A4 = -4.7192E-02$
 $-7.0688E-02$ $3.9453E-02$ $-2.0570E-02$ $1.8352E-01$ $3.3300E-02$ $A6 = 1.3599E-02$ $6.7547E-02$
 $-4.7790E-02$ $-1.9058E-01$ $-4.1269E-01$ $-7.7319E-02$ $A8 = -1.8138E-02$ $-9.4969E-02$
 $-1.5747E-02$ $2.3811E-01$ $5.0227E-01$ $8.0864E-02$ $A10 = -1.7872E-02$ $2.6867E-03$
 $-9.1459E-02$ $-2.4357E-01$ $-3.4197E-02$ $A12 = 2.7318E-02$ $3.8233E-04$ Surface # 8 9 10 11 12
13 $k = -9.0249E+00$ $-1.8336E+01$ $-3.9819E+00$ $-3.0957E+00$ $1.9367E+00$ $-6.9866E+00$ $A4 =$
 $-9.8090E-02$ $-1.0549E-01$ $-2.6887E-01$ $-1.1386E-01$ $-5.3755E-02$ $-6.2835E-02$ $A6 =$
 $3.2554E-02$ $5.8699E-02$ $1.3592E-01$ $-9.7432E-02$ $2.7217E-02$ $2.6537E-02$ $A8 = -3.1922E-02$
 $-5.4107E-02$ $1.9802E-01$ $3.0547E-01$ $-1.1348E-02$ $-8.0041E-03$ $A10 = -3.2857E-02$
 $4.0901E-03$ $-2.2670E-01$ $-2.0319E-01$ $2.1500E-03$ $1.3291E-03$ $A12 = 1.7673E-02$ $7.7777E-03$
 $8.5227E-02$ $5.6128E-02$ $-1.5555E-04$ $-1.2193E-04$ $A14 = -1.3520E-03$ $-1.1376E-02$
 $-5.7523E-03$ $2.0108E-06$ $5.0552E-06$

(134) In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in Table 14 below are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided

again.

(135) Moreover, these parameters can be calculated from Table 12 and Table 13 and satisfy the conditions stated in Table 14.

(136) TABLE-US-00014 TABLE 14 Embodiment 5 f [mm] 4.13 Fno 2.20 HFOV [deg.] 36.4 V5/(V3 + V4) 1.19 CT2/(T12 + T23) 3.04 CT5/CT6 1.27 CT1/|Sag12| 1.21 |R1/f| 0.36 f/R3 2.21 R12/f 0.23 |f/f2| + |f/f3| 2.90 |f/f5| + |f/f6| 3.84

6th Embodiment

(137) FIG. 6A is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure. FIG. 6B shows, in order from left to right, longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment.

(138) In FIG. 6A, the image capturing device includes an image capturing optical lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 690. The image capturing optical lens assembly includes, in order from an object side to an image side, a first lens element 610, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, and a sixth lens element 660, wherein the image capturing optical lens assembly has a total of six lens elements (610-660) with refractive power.

(139) The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 610 is made of plastic material.

(140) The second lens element 620 with positive refractive power has an object-side surface 621 being convex in a paraxial region thereof and an image-side surface 622 being convex in a paraxial region thereof, which are both aspheric, and the second lens element 620 is made of plastic material.

(141) The third lens element 630 with negative refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being concave in a paraxial region thereof, which are both aspheric, and the third lens element 630 is made of plastic material. Furthermore, the object-side surface 631 of the third lens element 630 has at least one inflection point.

(142) The fourth lens element 640 with positive refractive power has an object-side surface 641 being convex in a paraxial region thereof and an image-side surface 642 being concave in a paraxial region thereof, which are both aspheric, and the fourth lens element 640 is made of plastic material. Furthermore, the image-side surface 642 of the fourth lens element 640 has at least one convex shape in an off-axis region thereof.

(143) The fifth lens element 650 with positive refractive power has an object-side surface 651 being concave in a paraxial region thereof and an image-side surface 652 being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element 650 is made of plastic material.

(144) The sixth lens element 660 with negative refractive power has an object-side surface 661 being concave in a paraxial region thereof and an image-side surface 662 being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element 660 is made of plastic material. Furthermore, the image-side surface 662 of the sixth lens element 660 has at least one convex shape in an off-axis region thereof.

(145) The image capturing optical lens assembly further includes an aperture stop 600 and an IR-cut filter 670. The aperture stop 600 is disposed between the first lens element 610 and the second lens element 620. The IR-cut filter 670 is made of glass and located between the sixth lens element 660 and an image surface 680, and will not affect the focal length of the image capturing optical lens assembly.

(146) The image sensor 690 is disposed on or near the image surface 680 of the image capturing optical lens assembly.

(147) The detailed optical data of the sixth embodiment are shown in TABLE 15, and the aspheric surface data are shown in TABLE 16, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximal field of view.

(148) TABLE-US-00015 TABLE 15 (Embodiment 6) $f = 4.03$ mm, $Fno = 2.15$, $HFOV = 37.5$ deg.
Curvature Focal Surface # Radius Thickness Material Index Abbe # Length 0 Object Plano Infinity
1 Lens 1 1.700 ASP 0.335 Plastic 1.544 55.9 20.58 2 1.865 ASP 0.303 3 Ape. Stop Plano -0.047 4
Lens 2 1.901 ASP 0.534 Plastic 1.544 55.9 2.90 5 -8.295 ASP 0.050 6 Lens 3 102.239 ASP 0.265
Plastic 1.639 23.5 -4.00 7 2.492 ASP 0.403 8 Lens 4 4.782 ASP 0.333 Plastic 1.639 23.5 33.21 9
6.004 ASP 0.314 10 Lens 5 -3.233 ASP 0.678 Plastic 1.544 55.9 1.98 11 -0.866 ASP 0.050 12
Lens 6 -8.125 ASP 0.611 Plastic 1.544 55.9 -1.88 13 1.199 ASP 0.750 14 IR-cut filter Plano 0.300
Glass 1.517 64.2 15 Plano 0.311 16 Image Surface Plano — Note: Reference wavelength is d-line
587.6 nm

(149) TABLE-US-00016 TABLE 16 Aspheric Coefficients Surface # 1 2 4 5 6 7 $k = -3.2057E-01$
 $-5.8285E-01$ $-4.5187E+00$ $-9.0000E+01$ $-9.0000E+01$ $2.9881E+00$ $A4 = -2.6126E-02$
 $-5.5272E-02$ $4.9583E-02$ $9.0479E-02$ $9.2170E-02$ $-2.3305E-02$ $A6 = -6.0019E-03$
 $-3.1834E-02$ $-4.9241E-02$ $-3.0985E-01$ $-3.0386E-01$ $-4.1487E-02$ $A8 = -1.2882E-02$
 $3.1790E-03$ $-1.8385E-02$ $3.3010E-01$ $3.9526E-01$ $6.9733E-02$ $A10 = 6.7825E-03$ $1.6460E-02$
 $-1.3335E-01$ $-1.7932E-01$ $-4.0368E-02$ $A12 = -3.9320E-03$ $5.2719E-03$ $-1.1145E-03$ Surface
8 9 10 11 12 13 $k = 1.1940E+01$ $-2.0465E+01$ $-1.6368E+00$ $-2.5106E+00$ $9.2904E+00$
 $-8.8205E+00$ $A4 = -7.1430E-02$ $-5.6625E-02$ $-1.0770E-01$ $2.4797E-02$ $-1.4956E-02$
 $-5.4797E-02$ $A6 = -8.5848E-03$ $3.2675E-02$ $-1.3003E-02$ $-1.9988E-01$ $3.9582E-02$
 $2.2760E-02$ $A8 = -1.5609E-02$ $-5.6250E-02$ $1.7688E-01$ $2.7655E-01$ $-2.2307E-02$
 $-7.0516E-03$ $A10 = -3.1221E-03$ $3.9067E-02$ $-1.3908E-01$ $-1.4230E-01$ $5.6135E-03$
 $1.2827E-03$ $A12 = 4.7077E-04$ $-1.7340E-02$ $4.3536E-02$ $3.2856E-02$ $-6.5853E-04$
 $-1.4239E-04$ $A14 = 3.5450E-03$ $-5.1505E-03$ $-2.8977E-03$ $3.0064E-05$ $7.2813E-06$

(150) In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in Table 17 below are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

(151) Moreover, these parameters can be calculated from Table 15 and Table 16 and satisfy the conditions stated in Table 17.

(152) TABLE-US-00017 TABLE 17 6.sup.th Embodiment f [mm] 4.03 Fno 2.15 $HFOV$ [deg.] 37.5
 $V5/(V3 + V4)$ 1.19 $CT2/(T12 + T23)$ 1.75 $CT5/CT6$ 1.11 $CT1/|Sag12|$ 1.76 $|R1/f|$ 0.42 $f/R3$ 2.12
 $R12/f$ 0.30 $|f/f2| + |f/f3|$ 2.40 $|f/f5| + |f/f6|$ 4.18

7th Embodiment

(153) FIG. 7A is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure. FIG. 7B shows, in order from left to right, longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment.

(154) In FIG. 7A, the image capturing device includes an image capturing optical lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 790. The image capturing optical lens assembly includes, in order from an object side to an image side, a first lens element 710, a second lens element 720, a third lens element 730, a fourth lens element 740, a fifth lens element 750, and a sixth lens element 760, wherein the image capturing optical lens assembly has a total of six lens elements (710-760) with refractive power.

(155) The first lens element 710 with positive refractive power has an object-side surface 711 being convex in a paraxial region thereof and an image-side surface 712 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 710 is made of plastic material.

(156) The second lens element 720 with positive refractive power has an object-side surface 721

being convex in a paraxial region thereof and an image-side surface **722** being convex in a paraxial region thereof, which are both aspheric, and the second lens element **720** is made of plastic material.

(157) The third lens element **730** with negative refractive power has an object-side surface **731** being concave in a paraxial region thereof and an image-side surface **732** being concave in a paraxial region thereof, which are both aspheric, and the third lens element **730** is made of plastic material. Furthermore, the object-side surface **731** of the third lens element **730** has at least one inflection point.

(158) The fourth lens element **740** with positive refractive power has an object-side surface **741** being convex in a paraxial region thereof and an image-side surface **742** being concave in a paraxial region thereof, which are both aspheric, and the fourth lens element **740** is made of plastic material. Furthermore, the image-side surface **742** of the fourth lens element **740** has at least one convex shape in an off-axis region thereof.

(159) The fifth lens element **750** with positive refractive power has an object-side surface **751** being concave in a paraxial region thereof and an image-side surface **752** being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element **750** is made of plastic material.

(160) The sixth lens element **760** with negative refractive power has an object-side surface **761** being convex in a paraxial region thereof and an image-side surface **762** being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element **760** is made of plastic material.

(161) The image capturing optical lens assembly further includes an aperture stop **700** and an IR-cut filter **770**. The aperture stop **700** is disposed between an imaged object and the first lens element **710**. The IR-cut filter **770** is made of glass and located between the sixth lens element **760** and an image surface **780**, and will not affect the focal length of the image capturing optical lens assembly.

(162) The image sensor **790** is disposed on or near the image surface **780** of the image capturing optical lens assembly.

(163) The detailed optical data of the seventh embodiment are shown in TABLE 18, and the aspheric surface data are shown in TABLE 19, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximal field of view.

(164) TABLE-US-00018
TABLE 18 (Embodiment 7) f = 3.68 mm, Fno = 2.08, HFOV = 39.3 deg.
Curvature Focal Surface # Radius Thickness Material Index Abbe # Length 0 Object Plano Infinity
1 Ape. Stop Plano -0.173 2 Lens 1 1.899 ASP 0.300 Plastic 1.544 55.9 122.76 3 1.846 ASP 0.116 4
Lens 2 2.300 ASP 0.554 Plastic 1.544 55.9 2.78 5 -4.035 ASP 0.081 6 Lens 3 -38.337 ASP 0.240
Plastic 1.633 23.4 -4.10 7 2.790 ASP 0.286 8 Lens 4 3.425 ASP 0.258 Plastic 1.633 23.4 1256.25 9
3.340 ASP 0.364 10 Lens 5 -9.692 ASP 0.867 Plastic 1.544 55.9 1.85 11 -0.942 ASP 0.050 12
Lens 6 2.933 ASP 0.458 Plastic 1.530 55.8 -1.94 13 0.720 ASP 0.750 14 IR-cut filter Plano 0.210
Glass 1.517 64.2 15 Plano 0.517 16 Image Surface Plano — Note: Reference wavelength is d-line
587.6 nm

(165) TABLE-US-00019
TABLE 19 Aspheric Coefficients Surface # 2 3 4 5 6 7 k= -9.5669E-01
-2.2189E+00 -9.9136E+00 -5.9386E+01 9.0000E+01 3.4245E+00 A4= -3.9422E-02
-9.3177E-02 1.0236E-02 -1.3459E-02 1.2391E-01 -8.7816E-03 A6= 3.6432E-03 2.3632E-02
-6.0008E-02 -1.9883E-01 -4.3066E-01 -1.2269E-01 A8= -5.6239E-03 -6.4246E-02
-3.4128E-02 1.7984E-01 4.4249E-01 1.0769E-01 A10= -1.5854E-02 1.5019E-02 1.8394E-02
-5.9817E-02 -2.0518E-01 -5.6384E-02 A12= 1.5100E-04 -7.1739E-04 -3.5962E-04
3.8510E-04 4.0032E-02 1.0816E-02 A14= -2.9618E-12 4.2165E-12 -8.4906E-04 3.1939E-04
5.3024E-05 5.6840E-06 A16= -3.0325E-13 5.8639E-13 -1.3039E-12 2.4245E-12 1.5149E-04
-2.1479E-05 Surface # 8 9 10 11 12 13 k= -4.7361E+01 -2.0709E+01 -4.6097E+00
-5.6391E+00 1.4561E-01 -4.5463E+00 A4= -5.3418E-02 -9.5577E-02 7.8529E-02

-1.0613E-01 -2.1438E-01 -8.9370E-02 A6= -2.5685E-02 2.3691E-02 -1.2547E-01
1.2850E-01 7.8166E-02 3.4161E-02 A8= 6.2631E-03 -1.9846E-02 1.3612E-01 -1.2213E-01
-2.2919E-02 -1.0138E-02 A10= 1.1778E-02 -1.2449E-02 -9.6794E-02 9.4158E-02
8.2159E-03 1.9611E-03 A12= -5.2422E-03 2.2092E-02 3.6505E-02 -4.0036E-02
-2.2675E-03 -2.3819E-04 A14= -2.5121E-06 -5.6189E-03 -5.4061E-03 8.2931E-03
3.2862E-04 1.6232E-05 A16= -6.5874E-06 -9.2016E-06 -1.4484E-04 -6.6813E-04
-1.8730E-05 -4.4817E-07

(166) In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in Table 20 below are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

(167) Moreover, these parameters can be calculated from Table 18 and Table 19 and satisfy the conditions stated in Table 20.

(168) TABLE-US-00020 TABLE 20 7.sup.th Embodiment f [mm] 3.68 Fno 2.08 HFOV [deg.] 39.3
V5/(V3 + V4) 1.19 CT2/(T12 + T23) 2.81 CT5/CT6 1.89 CT1/|Sag12| 2.23 |R1/f| 0.52 f/R3 1.60
R12/f 0.20 |f/f2| + |f/f3| 2.22 |f/f5| + |f/f6| 3.89

8th Embodiment

(169) FIG. 8A is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure. FIG. 8B shows, in order from left to right, longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment.

(170) In FIG. 8A, the image capturing device includes an image capturing optical lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor **890**. The image capturing optical lens assembly includes, in order from an object side to an image side, a first lens element **810**, a second lens element **820**, a third lens element **830**, a fourth lens element **840**, a fifth lens element **850**, and a sixth lens element **860**, wherein the image capturing optical lens assembly has a total of six lens elements (**810-860**) with refractive power.

(171) The first lens element **810** with negative refractive power has an object-side surface **811** being convex in a paraxial region thereof and an image-side surface **812** being concave in a paraxial region thereof, which are both aspheric, and the first lens element **810** is made of plastic material.

(172) The second lens element **820** with positive refractive power has an object-side surface **821** being convex in a paraxial region thereof and an image-side surface **822** being concave in a paraxial region thereof, which are both aspheric, and the second lens element **820** is made of plastic material.

(173) The third lens element **830** with negative refractive power has an object-side surface **831** being concave in a paraxial region thereof and an image-side surface **832** being concave in a paraxial region thereof, which are both aspheric, and the third lens element **830** is made of plastic material. Furthermore, the object-side surface **831** of the third lens element **830** has at least one inflection point.

(174) The fourth lens element **840** with positive refractive power has an object-side surface **841** being convex in a paraxial region thereof and an image-side surface **842** being concave in a paraxial region thereof, which are both aspheric, and the fourth lens element **840** is made of plastic material. Furthermore, the image-side surface **842** of the fourth lens element **840** has at least one convex shape in an off-axis region thereof.

(175) The fifth lens element **850** with positive refractive power has an object-side surface **851** being concave in a paraxial region thereof and an image-side surface **852** being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element **850** is made of plastic material.

(176) The sixth lens element **860** with negative refractive power has an object-side surface **861** being convex in a paraxial region thereof and an image-side surface **862** being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element **860** is made of plastic material. Furthermore, the image-side surface **862** of the sixth lens element **860** has at least one convex shape in an off-axis region thereof.

(177) The image capturing optical lens assembly further includes an aperture stop **800** and an IR-cut filter **870**. The aperture stop **800** is disposed between an imaged object and the first lens element **810**. The IR-cut filter **870** is made of glass and located between the sixth lens element **860** and an image surface **880**, and will not affect the focal length of the image capturing optical lens assembly.

(178) The image sensor **890** is disposed on or near the image surface **880** of the image capturing optical lens assembly.

(179) The detailed optical data of the eighth embodiment are shown in TABLE 21, and the aspheric surface data are shown in TABLE 22, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximal field of view.

(180) TABLE-US-00021 TABLE 21 (Embodiment 8) $f = 3.92$ mm, $Fno = 2.28$, $HFOV = 38.2$ deg.
Curvature Focal Surface # Radius Thickness Material Index Abbe # Length 0 Object Plano Infinity
1 Ape. Stop Plano -0.145 2 Lens 1 2.105 ASP 0.300 Plastic 1.544 55.9 -37.93 3 1.814 ASP 0.078
4 Lens 2 1.550 ASP 0.437 Plastic 1.544 55.9 2.89 5 100.000 ASP 0.114 6 Lens 3 -17.344 ASP
 0.240 Plastic 1.640 23.5 -4.67 7 3.634 ASP 0.228 8 Lens 4 3.479 ASP 0.422 Plastic 1.544 55.9
 17.79 9 5.198 ASP 0.348 10 Lens 5 -39.154 ASP 1.084 Plastic 1.544 55.9 2.32 11 -1.235 ASP
 0.050 12 Lens 6 3.049 ASP 0.457 Plastic 1.544 55.9 -2.25 13 0.828 ASP 0.500 14 IR-cut filter
Plano 0.210 Glass 1.517 64.2 15 Plano 0.724 16 Image Surface -66.000 — Note: Reference
wavelength is d-line 587.6 nm

(181) TABLE-US-00022 TABLE 22 Aspheric Coefficients Surface # 2 3 4 5 6 7 $k = -1.4899E+00$
 $-9.8839E+00$ $-6.8085E+00$ $9.0000E+01$ $-9.0000E+01$ $6.1144E+00$ $A4 = -4.5205E-02$
 $-1.2247E-01$ $-3.2748E-02$ $4.9659E-02$ $1.3790E-01$ $5.3820E-02$ $A6 = 5.1197E-02$ $6.0345E-02$
 $-7.3451E-02$ $-2.4952E-01$ $-3.3291E-01$ $-1.0592E-01$ $A8 = -4.0943E-02$ $-6.6411E-02$
 $1.2284E-03$ $2.1260E-01$ $4.2570E-01$ $1.1785E-01$ $A10 = -8.6014E-03$ $-5.0626E-03$
 $-3.0791E-02$ $-8.5375E-02$ $-2.3612E-01$ $-5.9685E-02$ $A12 = 1.5100E-04$ $-7.1739E-04$
 $-3.5962E-04$ $3.8509E-04$ $4.7718E-02$ $1.1319E-02$ $A14 = -1.2201E-12$ $7.0003E-12$
 $-8.4906E-04$ $3.1939E-04$ $5.3133E-05$ $5.2256E-06$ $A16 = -2.2709E-13$ $9.4712E-13$
 $-2.7109E-12$ $4.6755E-12$ $1.5149E-04$ $-2.1479E-05$ Surface # 8 9 10 11 12 13 $k = -4.6730E+01$
 $-1.5373E+01$ $9.0000E+01$ $-7.1106E+00$ $4.9048E-01$ $-4.5643E+00$ $A4 = -5.4516E-03$
 $-1.0078E-01$ $4.2271E-02$ $-2.3773E-02$ $-2.2898E-01$ $-1.1128E-01$ $A6 = -3.1108E-02$
 $3.0750E-02$ $-1.2529E-01$ $-1.9586E-02$ $6.3251E-02$ $5.3012E-02$ $A8 = 6.8351E-03$ $-1.5314E-02$
 $1.3527E-01$ $3.1412E-02$ $2.0475E-03$ $-1.9274E-02$ $A10 = 1.7495E-02$ $-1.4654E-02$
 $-9.7150E-02$ $-9.8738E-03$ $-5.6386E-03$ $4.7017E-03$ $A12 = -5.2383E-03$ $2.1011E-02$
 $3.6467E-02$ $2.4407E-04$ $1.4705E-03$ $-7.3304E-04$ $A14 = -2.7941E-06$ $-5.2369E-03$
 $-5.2777E-03$ $3.4269E-04$ $-1.6603E-04$ $6.4713E-05$ $A16 = -6.5874E-06$ $-1.0560E-05$
 $-1.6489E-04$ $-4.3067E-05$ $6.9222E-06$ $-2.4059E-06$

(182) In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in Table 23 below are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

(183) Moreover, these parameters can be calculated from Table 21 and Table 22 and satisfy the conditions stated in Table 23.

(184) TABLE-US-00023 TABLE 23 8.sup.th Embodiment f [mm] 3.92 Fno 2.28 $HFOV$ [deg.] 38.2
 $V5/(V3 + V4)$ 0.70 $CT2/(T12 + T23)$ 2.28 $CT5/CT6$ 2.37 $CT1/|Sag12|$ 3.96 $|R1/f|$ 0.54 $f/R3$ 2.53

$$R12/f\ 0.21\ |f/f2| + |f/f3|\ 2.20\ |f/f5| + |f/f6|\ 3.43$$

9th Embodiment

(185) FIG. 9A is a schematic view of an image capturing device according to the 9th embodiment of the present disclosure. FIG. 9B shows, in order from left to right, longitudinal spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 9th embodiment.

(186) In FIG. 9A, the image capturing device includes an image capturing optical lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 990. The image capturing optical lens assembly includes, in order from an object side to an image side, a first lens element 910, a second lens element 920, a third lens element 930, a fourth lens element 940, a fifth lens element 950, and a sixth lens element 960, wherein the image capturing optical lens assembly has a total of six lens elements (910-960) with refractive power.

(187) The first lens element 910 with negative refractive power has an object-side surface 911 being convex in a paraxial region thereof and an image-side surface 912 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 910 is made of plastic material.

(188) The second lens element 920 with positive refractive power has an object-side surface 921 being convex in a paraxial region thereof and an image-side surface 922 being convex in a paraxial region thereof, which are both aspheric, and the second lens element 920 is made of plastic material.

(189) The third lens element 930 with negative refractive power has an object-side surface 931 being convex in a paraxial region thereof and an image-side surface 932 being concave in a paraxial region thereof, which are both aspheric, and the third lens element 930 is made of plastic material. Furthermore, the object-side surface 931 of the third lens element 930 has at least one inflection point.

(190) The fourth lens element 940 with negative refractive power has an object-side surface 941 being convex in a paraxial region thereof and an image-side surface 942 being concave in a paraxial region thereof, which are both aspheric, and the fourth lens element 940 is made of plastic material. Furthermore, the image-side surface 942 of the fourth lens element 940 has at least one convex shape in an off-axis region thereof.

(191) The fifth lens element 950 with positive refractive power has an object-side surface 951 being plano in a paraxial region thereof and an image-side surface 952 being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element 950 is made of plastic material.

(192) The sixth lens element 960 with negative refractive power has an object-side surface 961 being convex in a paraxial region thereof and an image-side surface 962 being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element 960 is made of plastic material. Furthermore, the image-side surface 962 of the sixth lens element 960 has at least one convex shape in an off-axis region thereof.

(193) The image capturing optical lens assembly further includes an aperture stop 900 and an IR-cut filter 970. The aperture stop 900 is disposed between an imaged object and the first lens element 910. The IR-cut filter 970 is made of glass and located between the sixth lens element 960 and an image surface 980, and will not affect the focal length of the image capturing optical lens assembly.

(194) The image sensor 990 is disposed on or near the image surface 980 of the image capturing optical lens assembly.

(195) The detailed optical data of the ninth embodiment are shown in TABLE 24, and the aspheric surface data are shown in TABLE 25, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximal field of view.

(196) TABLE-US-00024 TABLE 24 (Embodiment 9) f = 3.62 mm, Fno = 1.98, HFOV = 40.2 deg.
Curvature Focal Surface # Radius Thickness Material Index Abbe # Length 0 Object Plano Infinity

1 Ape. Stop Plano -0.180 2 Lens 1 1.747 ASP 0.260 Plastic 1.633 23.4 -17.23 3 1.419 ASP 0.117 4 Lens 2 2.225 ASP 0.622 Plastic 1.544 55.9 2.86 5 -4.663 ASP 0.088 6 Lens 3 4.938 ASP 0.250 Plastic 1.633 23.4 -7.00 7 2.289 ASP 0.306 8 Lens 4 5.050 ASP 0.325 Plastic 1.583 30.2 -22.24 9 3.549 ASP 0.389 10 Lens 5 ∞ ASP 0.729 Plastic 1.544 55.9 2.13 11 -1.157 ASP 0.053 12 Lens 6 1.924 ASP 0.449 Plastic 1.544 55.9 -2.51 13 0.733 ASP 0.800 14 IR-cut filter Plano 0.210 Glass 1.517 64.2 15 Plano 0.485 16 Image Surface Plano — Note: Reference wavelength is d-line 587.6 nm

(197) TABLE-US-00025 TABLE 25 Aspheric Coefficients Surface # 2 3 4 5 6 7 k= -1.5406E+00 -1.4741E+00 -6.9574E+00 -9.0000E+01 -7.7354E+01 1.4291E+00 A4= -5.9518E-02 -8.4009E-02 3.8285E-02 -7.5145E-02 1.0508E-01 -2.5999E-02 A6= 5.0380E-03 1.3010E-04 -7.6756E-02 -1.0269E-01 -4.1895E-01 -1.4113E-01 A8= -2.9733E-03 1.3313E-02 4.9236E-02 1.3593E-01 4.4191E-01 1.2966E-01 A10= -1.3711E-02 -2.4812E-02 -8.7498E-03 -3.4919E-02 -1.8266E-01 -4.2303E-02 A12= -4.7003E-06 -1.8784E-06 -7.1207E-07 1.4858E-06 2.0534E-02 8.0679E-04 A14= -4.8754E-07 -1.2373E-06 4.9004E-07 2.0374E-06 -3.2960E-07 -1.5009E-04 A16= -3.8986E-08 -2.5634E-08 -2.4542E-07 5.7090E-07 -6.8212E-07 4.5550E-07 Surface # 8 9 10 11 12 13 k= -9.0000E+01 -7.7593E+00 0.0000E+00 -5.9615E+00 -4.6525E-01 -3.6454E+00 A4= -7.7485E-02 -1.4523E-01 9.3359E-02 -9.2801E-03 -2.5153E-01 -1.1884E-01 A6= -2.2953E-02 2.1720E-02 -8.8561E-02 2.0840E-02 6.2415E-02 5.1561E-02 A8= 1.1669E-01 5.1752E-02 4.2159E-02 -1.7276E-02 -5.0257E-03 -1.6872E-02 A10= -1.6451E-01 -1.0054E-01 -4.9625E-03 2.2422E-02 2.1098E-03 3.8037E-03 A12= 1.3794E-01 7.7829E-02 -8.1567E-03 -1.1801E-02 -1.3193E-03 -5.5778E-04 A14= -5.4488E-02 -2.4154E-02 4.6838E-03 2.4895E-03 2.6217E-04 4.6576E-05 A16= 7.4997E-03 2.3888E-03 -8.5860E-04 -1.8221E-04 -1.7122E-05 -1.6371E-06

(198) In the 9th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in Table 26 below are the same as those stated in the 1st embodiment with corresponding values for the 9th embodiment, so an explanation in this regard will not be provided again.

(199) Moreover, these parameters can be calculated from Table 24 and Table 25 and satisfy the conditions stated in Table 26.

(200) TABLE-US-00026 TABLE 26 9.sup.th Embodiment f [mm] 3.62 Fno 1.98 HFOV [deg.] 40.2 V5/(V3 + V4) 1.04 CT2/(T12 + T23) 3.03 CT5/CT6 1.62 CT1/|Sag12| 1.08 |R1/f| 0.48 f/R3 1.63 R12/f 0.20 |f/f2| + |f/f3| 1.78 |f/f5| + |f/f6| 3.14

(201) The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-26 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

Claims

1. An image capturing optical lens assembly comprising in order from an object side to an image side: a first lens element having an image-side surface being concave in a paraxial region thereof; a second lens element; a third lens element; a fourth lens element having an image-side surface being concave in a paraxial region thereof; a fifth lens element with positive refractive power; and a sixth

lens element having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, a curvature radius of an object-side surface of the first lens element and a curvature radius of an image-side surface of the third lens element have the same sign, an absolute value of a curvature radius of an object-side surface of the second lens element is smaller than an absolute value of a curvature radius of an object-side surface of the fourth lens element, an axial distance between the fifth lens element and the sixth lens element is smaller than an axial distance between the first lens element and the second lens element; wherein a total number of lens elements having refractive power of the image capturing optical lens assembly is six, an f-number of the image capturing optical lens assembly is F_{no} , a central thickness of the fifth lens element is CT_5 , a central thickness of the sixth lens element is CT_6 , and the following conditions are satisfied:

$1.6 < F_{no} \leq 2.28$; and

$1.62 \leq CT_5/CT_6$.

2. The image capturing optical lens assembly of claim 1, wherein the second lens element has positive refractive power and an object-side surface being convex in a paraxial region thereof, the fifth lens element has an image-side surface being convex in a paraxial region thereof, and an aperture stop is positioned disposed between an imaged object and the third lens element.

3. The image capturing optical lens assembly of claim 1, wherein the first lens element has negative refractive power and the sixth lens element has negative refractive power.

4. The image capturing optical lens assembly of claim 1, wherein a focal length of the image capturing optical lens assembly is f , a focal length of the fifth lens element is f_5 , a focal length of the sixth lens element is f_6 , and the following condition is satisfied:

$1.50 < |f/f_5| + |f/f_6|$.

5. The image capturing optical lens assembly of claim 1, wherein a curvature radius of the image-side surface of the sixth lens element is R_{12} , a focal length of the image capturing optical lens assembly is f , and the following condition is satisfied:

$0 < R_{12}/f < 0.40$.

6. The image capturing optical lens assembly of claim 1, wherein a central thickness of the first lens element is CT_1 , a distance in parallel with an optical axis from an axial vertex on the image-side surface of the first lens element to a maximum effective radius position on the image-side surface of the first lens element is Sag_{12} , the central thickness of the fifth lens element is CT_5 , the central thickness of the sixth lens element is CT_6 , and the following conditions are satisfied:

$CT_1/|Sag_{12}| < 2.5$; and

$1.89 \leq CT_5/CT_6$.

7. The image capturing optical lens assembly of claim 1, wherein an absolute value of a focal length of the third lens element is smaller than an absolute value of a focal length of the fourth lens element.

8. The image capturing optical lens assembly of claim 1, wherein an axial distance between the first lens element and the second lens element is smaller than a central thickness of the first lens element.

9. The image capturing optical lens assembly of claim 1, wherein an absolute value of a curvature radius of the object-side surface of the sixth lens element is smaller than an absolute value of a curvature radius of an image-side surface of the second lens element.

10. The image capturing optical lens assembly of claim 1, wherein a focal length of the first lens element and a focal length of the third lens element have different signs.

11. The image capturing optical lens assembly of claim 1, wherein the axial distance between the fifth lens element and the sixth lens element is smaller than an axial distance between the second lens element and the third lens element.

12. An image capturing device, comprising the image capturing optical lens assembly of claim 1; and an image sensor, wherein the image sensor is disposed on an image surface of the image

capturing optical lens assembly.

13. An electronic device, comprising the image capturing device of claim 12.

14. An image capturing optical lens assembly comprising in order from an object side to an image side: a first lens element having negative refractive power and an image-side surface being concave in a paraxial region thereof; a second lens element; a third lens element; a fourth lens element; a fifth lens element; and a sixth lens element having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, an absolute value of a focal length of the third lens element is smaller than an absolute value of a focal length of the fourth lens element, an absolute value of a curvature radius of an image-side surface of the third lens element is smaller than an absolute value of a curvature radius of an object-side surface of the third lens element, an axial distance between the fifth lens element and the sixth lens element is smaller than an axial distance between the first lens element and the second lens element; wherein a total number of lens elements having refractive power of the image capturing optical lens assembly is six, an f-number of the image capturing optical lens assembly is F_{no} , a focal length of the image capturing optical lens assembly is f , a focal length of the fifth lens element is f_5 , a focal length of the sixth lens element is f_6 , and the following conditions are satisfied:

$1.6 < F_{no} \leq 2.28$; and

$1.50 < |f/f_5| + |f/f_6|$.

15. The image capturing optical lens assembly of claim 14, wherein the second lens element has positive refractive power, the fifth lens element has positive refractive power, the sixth lens element has negative refractive power, and an aperture stop is disposed between an imaged object and the third lens element.

16. The image capturing optical lens assembly of claim 14, wherein the second lens element has an object-side surface being convex in a paraxial region thereof and the fifth lens element has an image-side surface being convex in a paraxial region thereof.

17. The image capturing optical lens assembly of claim 14, wherein a curvature radius of the image-side surface of the sixth lens element is R_{12} , the focal length of the image capturing optical lens assembly is f , and the following condition is satisfied:

$0 < R_{12}/f < 0.40$.

18. The image capturing optical lens assembly of claim 14, wherein a central thickness of the fifth lens element is CT_5 , a central thickness of the sixth lens element is CT_6 , and the following condition is satisfied:

$0.80 < CT_5/CT_6$.

19. The image capturing optical lens assembly of claim 18, wherein the central thickness of the fifth lens element is CT_5 , the central thickness of the sixth lens element is CT_6 , and the following condition is satisfied:

$1.27 \leq CT_5/CT_6$.

20. The image capturing optical lens assembly of claim 14, wherein a central thickness of the first lens element is CT_1 , a distance in parallel with an optical axis from an axial vertex on the image-side surface of the first lens element to a maximum effective radius position on the image-side surface of the first lens element is Sag_{12} , and the following condition is satisfied:

$CT_1/|Sag_{12}| < 2.5$.

21. The image capturing optical lens assembly of claim 14, wherein the axial distance between the first lens element and the second lens element is smaller than a central thickness of the first lens element.

22. The image capturing optical lens assembly of claim 14, wherein an absolute value of a curvature radius of an object-side surface of the second lens element is smaller than an absolute value of a curvature radius of an object-side surface of the fourth lens element, and an absolute value of a curvature radius of the object-side surface of the sixth lens element is smaller than an absolute value of a curvature radius of an image-side surface of the second lens element.

23. The image capturing optical lens assembly of claim 14, wherein the fourth lens element has an image-side surface being concave in a paraxial region thereof.
24. An image capturing device, comprising the image capturing optical lens assembly of claim 14; and an image sensor, wherein the image sensor is disposed on an image surface of the image capturing optical lens assembly.
25. An electronic device, comprising the image capturing device of claim 24.
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